

SD Glossary

Terms used in SD Core documentation, from account to VOC.

This glossary defines the terms used throughout the SD Core for Windows documentation set. Terms are listed in alphabetical order.

A

Account — a workspace: a directory containing one or more files, a VOC, and its own `bp` source file. An account maps to a Windows group (`sdu_<name>` for user accounts, `sdg_<name>` for group accounts). Entry to an account is membership of its Windows group.

Administrator — the highest account tier. An administrator account is a member of the Windows `Administrators` group and receives the full VOC: 416 records. Administration is gated on elevation, not on a password.

Alternate key index — a secondary access path to records in a file, built from the values in a nominated field. Created with `create.index`, built with `build.index`.

Association — a group of dictionary fields that share multivalues. When fields are associated, a value mark in one corresponds to a value mark in all of them at the same position.

Attribute — a field in a dictionary record that defines one column of a query report. Dictionary field 1 is the name, field 2 is the type, field 3 is the conversion, field 4 is the correlative, field 5 is the format, field 6 is the header, field 7 is the association.

B

Background process — a process started with `phantom` that runs without a terminal. The process that starts it does not wait for it.

Break key — the interrupt key (Ctrl+C on Windows). Stops a running program or query and enters the debugger if one is active.

BASIC — the programming language. See SDBasic.

bp — the source file. An account's `bp` file is a directory file — an ordinary Windows folder with one file per program. `gp1.bp` is the shipped source for the system programs.

C

Catalogue — the registry of compiled programs. A catalogued program can be run by name from any account that can reach it. Private catalogue is per-account; local catalogue puts the VOC entry in the account's own VOC; global catalogue is system-wide and requires an administrator.

Common block — a named memory area shared between subroutines. Declared with `common` or `common /name/`. Used for passing data without arguments.

Conversion code — a dictionary field 3 entry that transforms data on input (`iconv`) or output (`oconv`). Examples: `D` for dates, `MR` for masked decimal, `MC` for case conversion.

Correlative — a dictionary field 4 entry that defines pre-processing of a field value before the conversion is applied. Examples: `A` (arithmetic), `F` (function), `I` (index).

D

Dictionary — a file that defines the fields of a data file for the query processor. Each record in a dictionary file describes one field. The dictionary file is named `<filename>.dict` and lives beside the data file.

Directory file — a file type where each record is a file on disk, stored in an ordinary Windows folder. Record ids are file names. The `bp` file is a directory file.

Dynamic array — a string containing field marks, value marks and subvalue marks. SDBasic's primary data structure for record manipulation. Functions: `extract`, `ins`, `del`, `replace`, `dcount`, `locate`.

Dynamic file — a file type stored in SD's own binary format, with a hash-based group structure and automatic resizing. Not readable by ordinary Windows programs.

F

Field mark — the delimiter (char 254) that separates fields in a dynamic array.

File — a collection of records, each identified by a record id. A file has a data portion and a dictionary portion. The two file types are directory files and dynamic files.

Format — dictionary field 5. Controls how a value is displayed: width, alignment, masking. The format specification syntax supports date masks, numeric masks, and text formatting.

F-pointer — a VOC entry of type `F` that points to another file, possibly in another account. A remote file pointer.

G

Group — (1) a Windows group used for account membership. (2) In a dynamic file, the bucket that holds records hashed to the same slot.

Group account — a shared workspace with no Windows account and no sign-in of its own. Created with `create.account group`. Reached with `logto` or through an F-pointer.

I

I-type — a dictionary field type whose value is computed by an SDBasic expression rather than stored in the record. The expression is compiled and stored in field 2 of the dictionary record.

Index — see Alternate key index.

L

List — see Select list.

Lock — a claim on a record or file that prevents other sessions from updating it simultaneously. Update locks (`readu`, `writeu`, `deleteu`) are exclusive; shared locks (`readl`) are not. Task locks are per-session and never block the same session.

M

Mark mapping — the translation of dynamic array marks to and from the separators used by sequential file I/O. Controlled per file with `SDBMarkMapping` in the API, or by the `setmark` and `setemark` statements in SDBasic.

Matrix — a two-dimensional array in SDBasic. Declared with `dimension` or `dim`. Indexed from 1.

Multivalue — a value within a field, separated by a value mark (char 253). A field can hold any number of multivalues; each multivalue can hold any number of subvalues.

O

Overflow — in a dynamic file, the condition where a group has more data than fits in its primary bucket and additional buckets are chained to it. Excessive overflow degrades performance and is reported by `analyse.file`.

P

p-code — the compiled bytecode that SD executes. Stored in the pcode library in `<sysdir>\bin` and loaded into shared memory at start-up.

Paragraph — a VOC entry of type `PA` that holds a sequence of TCL commands. Replaces the removed PROC language.

Phantom — see Background process.

Programmer — the middle account tier. Gets $82 + 42 = 124$ verbs: everything a standard account has, plus the development set (compile, catalogue, edit, file creation, index management, bulk record editing, process introspection).

Q

QMSYS — what the system account is called in OpenQM. In SD it is `SDSYS`, and `sdsys` on disk. Nothing in SD Core answers to the name QMSYS; it is here so that a reader coming from OpenQM finds the entry.

Query processor — the reporting language: `list`, `select`, `count`, `sort`, `sum`, and the rest. Reads dictionary definitions to format output. Invoked from TCL or from SDBasic with `select`, `readnext`, `getlist`.

R

Record — a single entry in a file, identified by a record id. A record is a dynamic array: a string of fields, values and subvalues.

Record id — the unique key of a record within a file. In a directory file, the record id is the file name.

S

SDBasic — the programming language of SD. A compiled BASIC with dynamic arrays, multivalued string functions, file I/O, select lists, sockets, sequential files, CSV handling, and transaction support.

Select list — an ordered list of record ids, built by `select`, `sselect`, `getlist`, or the API's `SDSelect`. Read with `readnext` or `SDReadNext`. Stored in the `$savedlists` file.

Sentence — a VOC entry of type `S` that holds a single TCL command with substitution parameters.

Session — one connection to SD, from sign-in to `off`. Each session runs as the invoking user's Windows identity.

Standard — the lowest account tier. Gets 82 verbs: enough to run an application and nothing that edits code or data in bulk. The default when no tier keyword is given on `create.account`.

Subvalue mark — the delimiter (char 252) that separates subvalues within a multivalue.

Suspended — a fourth account tier that denies all entry. Reversible with `modify.account <tier>`. Does not touch the VOC or Windows group membership.

T

TCL — the command processor. Reads what you type at the `:` prompt and dispatches it.

Terminfo — the terminal capability database. Defines what sequences a terminal sends for keys and what sequences to use for screen control. SD ships 63 definitions compiling to 100 terminal names. The terminfo compiler (`sdtic`) is not shipped with SD Core.

Transaction — a group of file updates that succeed or fail together. Declared with `start transaction` and committed with `commit` or rolled back with `rollback`.

V

Value mark — the delimiter (char 253) that separates multivalues within a field.

VOC — the Vocabulary file. Maps command names to the programs behind them. Each account has its own VOC. VOC entries are of types `V` (verb), `F` (file pointer), `PA` (paragraph), `S` (sentence), `K` (key), and `W` (external command).

W

Windows service — how SD runs on Windows. The service **String Database (SD)** is created by the installer, starts automatically at boot, and is removed by the uninstaller. Stopping it ends every session on the machine.
